Statistics for Computer Scientists

Lecture 4: Data collection

26 September 2025

Universiteit Leiden
The Netherlands

Where are we

Lecture	Topic	Book
5 Sept	Introduction, statistics fundamentals	Chapter 1
12 Sept	Descriptive statistics	Chapter 2
19 Sept	Bivariate relationships	Chapter 3
26 Sept	Data collection	Chapter 4
10 Oct	Sampling distributions	Chapter 7
17 Oct	Estimation	Chapter 8
31 Oct	Significance testing	Chapter 9
7 Nov	Comparing two groups	Chapter 10
14 Nov	Categorical data analysis	Chapter 11
21 Nov	Linear regression	Chapter 12
28 Nov	Multiple regression	Chapter 13
5 Dec	Tba	
12 Dec	Tba	

Today’s lecture

- Recap lecture 3

- Observational and experimental studies
- Data collection
- Sampling methods
- Bias

Recap

Universiteit Leiden
The Netherlands

Recap

- Categorical associations
  - Contingency table, conditional proportions
- Quantitative associations
  - Correlation, linear regression, scatter plot
- Prediction equation and interpretation
- Residuals

Observational and experimental studies

Universiteit Leiden
The Netherlands

Types of studies

- **Observational study**: values of explanatory and response variables are **observed** by the researcher

- **Experimental study**: researcher assigns subjects to a certain value of explanatory variables (different treatments)

Observational studies

- Remember: association does not imply causation!
- Observational: lurking variables can skew results

- Example: school drug testing
  - Two questions: “do you use drugs?” and “does your school test for drug use?”
  - Proportion of drug users similar for testing yes/no
  - So, does drug testing not affect drug use?

Experimental studies

- Create treatment groups as similar as possible
  - Same distribution for all variables that may be relevant!
- Makes it possible to draw causal conclusions*

- Disadvantages:
  - Ethics
  - Time and money
  - Monitoring

Big data

- Massive datasets collected by e.g. tech giants
- All observational data!
- Samples or populations?
- Inclusion of lurking variables

Data collection

Universiteit Leiden
The Netherlands

Data collection

- Also a statistician's task!

- Is the data of high enough quality?
- Is the sample reliable for the population of interest?
- Can we answer the research question with the collected data?

Example: predicting hemoglobin

- Blood bank tests hemoglobin (iron) level before blood donation
- Value must be ≥ 7.8 mmol/L for women to donate

- Research question: can we predict hemoglobin level based on sex, weight, and previous measurements?
  - E.g., using linear regression!

- Available data:
  - Hb distribution?

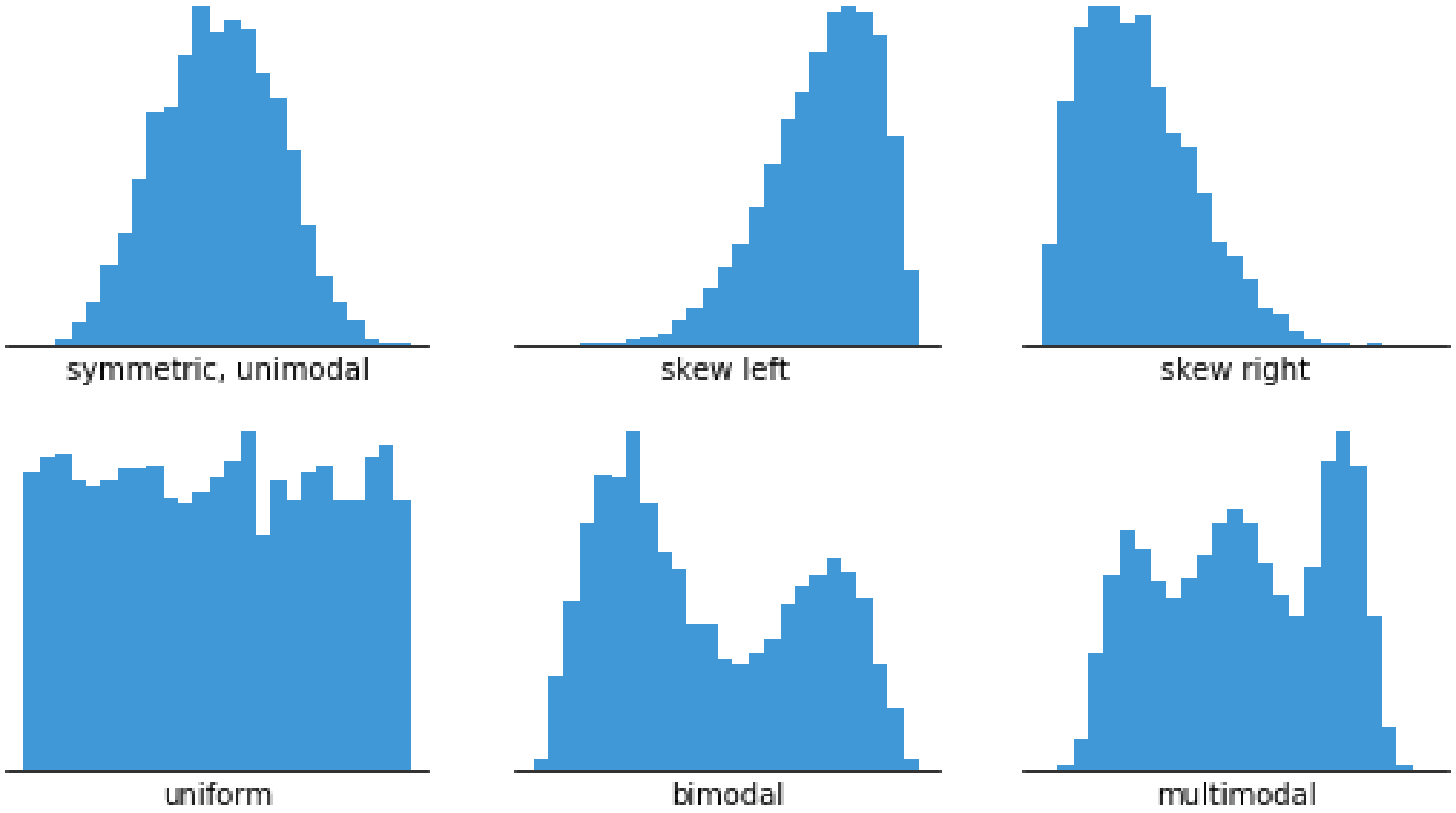
sex	weight	previous_Hb	days	Hb
F	70	8.3	140	8.5
F	55	7.6	180	7.8
F	100	8.0	122	8.1

Linear regression

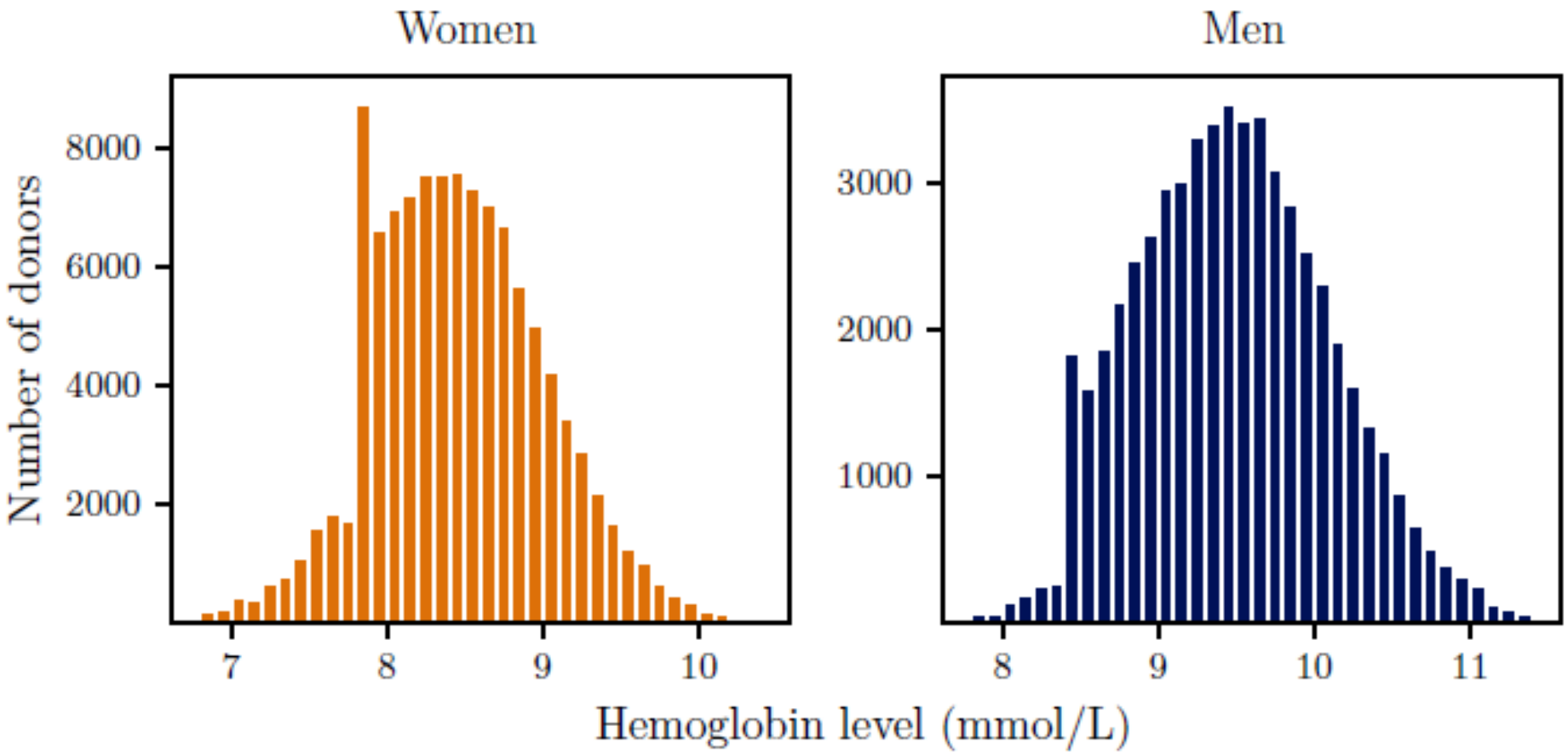
$$\hat{y} = a + b_1 * sex + b_2 * weight + b_3 * previous_{Hb} + b_4 * days$$

sex	weight	previous_Hb	days	Hb
F	70	8.3	140	8.5
F	55	7.6	180	7.8
F	100	8.0	122	8.1

Expected distribution of y?



Observed Hb distributions…



Data collection

- Data is collected as part of blood donation process, not for scientific purposes
- Measurement variability!
- Measuring Hb twice, take the difference between two measurements
  - Centered around zero: OK
  - Standard deviation ~0.6 mmol/L!

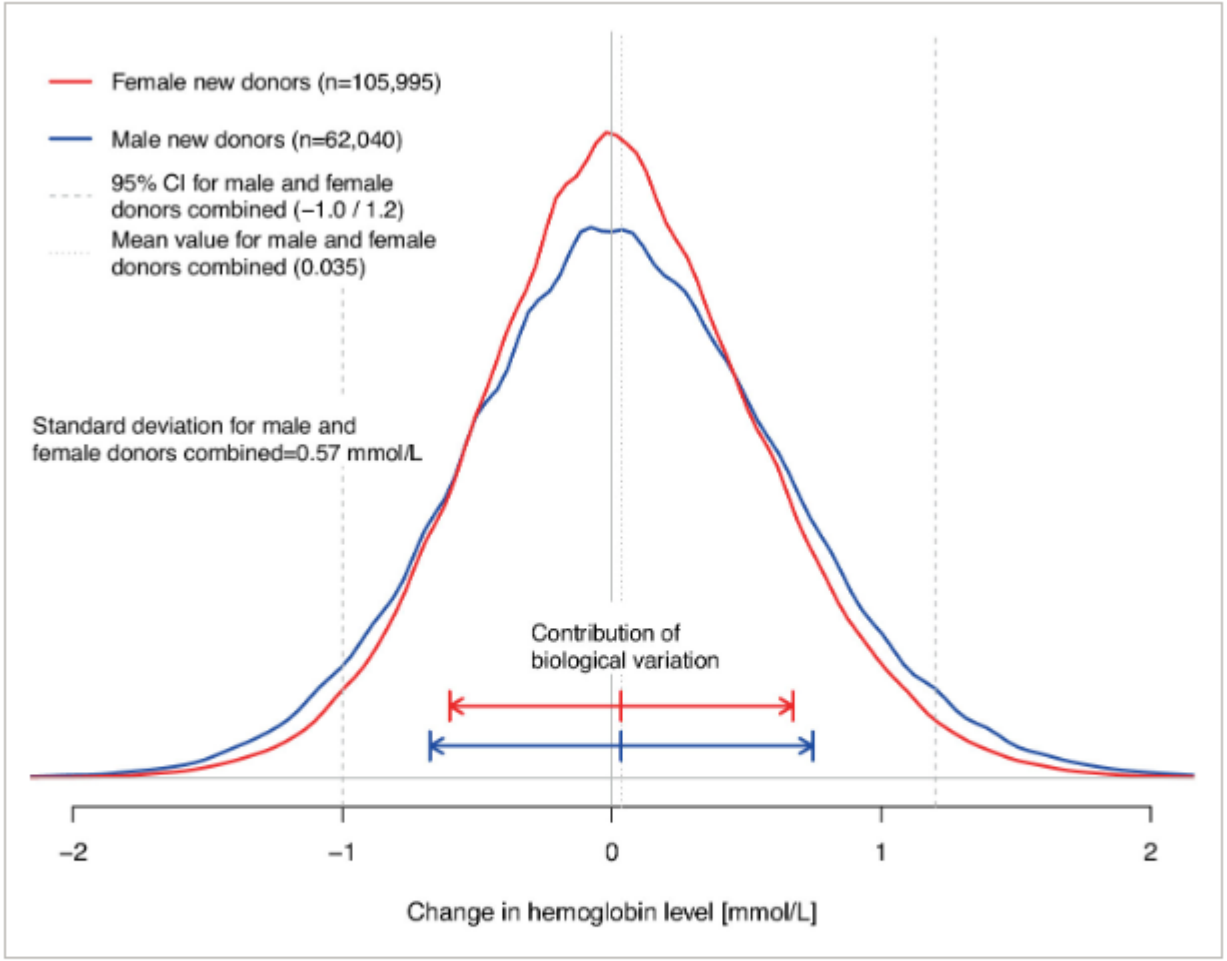


FIGURE 1 Open in figure viewer | PowerPoint

Distribution of the change in hemoglobin level measured between donor intake and first donation (within 42 days) of new donors. [Color figure can be viewed at wileyonlinelibrary.com]

Sampling methods

Universiteit Leiden
The Netherlands

How to sample?

1. Identify the subject
2. Compile the sampling frame
3. Choose a sampling design

Probability sampling

- **Probability sampling**: for every possible sample, we can specify the probability of it being selected.
  - *Example: every sample has the same probability*

- **Nonprobability sampling**: we **cannot** specify the probability of being selected for every possible sample.
  - *Example: students asking their friends to fill in a survey*

Methods of probability sampling

- Key mechanism: randomization

- Simple random sampling
- Systematic random sampling
- Stratified random sampling
- Cluster random sampling
- Multistage random sampling

Simple random sampling

- Every sample has the same probability of being selected

- Example: draw ballots from a hat, use a random number generator

```
000000000000000000000
000000000000000000000
000000000000000000000
000000000000000000000
000000000000000000000
```

Systematic random sampling

- Every k^{th} subject is selected for the sample

- Example: include every 10th person from an alphabetized list

```
0000000000000000000000
0000000000000000000000
0000000000000000000000
0000000000000000000000
0000000000000000000000
```

Stratified random sampling

- Divide the population into groups (strata) and select a simple random sample from each stratum.
- Options: proportional vs disproportional

- Use when groups are important, especially when small groups exist

- Example: select one house from every street in a municipality

Stratified random sampling

- Same size strata, proportional sampling

AAAAAAAAAAAAAAAAAAAAA
BBBBBBBBBBBBBBBBBBBBB
CCCCCCCCCCCCCCCCCCCCC
DDDDDDDDDDDDDDDDDDDDD
EEEEEEEEEEEEEEEEEEEEE

Stratified random sampling

- Different size strata, proportional sampling
- Proportions in population are same as proportions in sample

AAAAAAAAAAAAAAAAAAAAA
BBBBBBBBBBBBBBBBBBBBB
BBBBBBBBBBBBBBBBBBBBB
BBBBBBBBBBBBBBBBBBBBB
BBBBBBBBBBBBBBBBBBBBB

Stratified random sampling

- Different size strata, disproportional sampling
- Proportions in population differ from proportions in sample

AAAAAAAAAAAAAAAAAAAA
BBBBBBBBBBBBBBBBBBBB
BBBBBBBBBBBBBBBBBBBB
BBBBBBBBBBBBBBBBBBBB
BBBBBBBBBBBBBBBBBBBB

Cluster sampling

- Divide the population into groups (clusters) and select a simple random sample of clusters.

- Not all groups are included in the sample!
- Use when no group effect is hypothesized

- Example: select all houses in 10 streets of a municipality

Cluster sampling

- Divide the population into groups (clusters) and select a simple random sample of clusters.
- Not all groups are included in the sample!

AAAAAAAAAAAAAAAAAAAAA
BBBBBBBBBBBBBBBBBBBBB
CCCCCCCCCCCCCCCCCCCCC
DDDDDDDDDDDDDDDDDDDDD
EEEEEEEEEEEEEEEEEEEEE

Stratified vs cluster sampling

- Use stratified sampling to compare groups, or when differences between groups are expected

- Use cluster sampling when **no** difference between groups is expected, for a convenient sample

Multistage sampling

- Sampling is split over multiple ‘stages’

- Example: interested in male/female student happiness in the Netherlands

1. Select 5 universities (cluster sampling)
2. Select 10 study programmes per university (cluster sampling)
3. Select 50 female and 50 male students per study programme (stratified random sampling) (proportional or disproportional?)

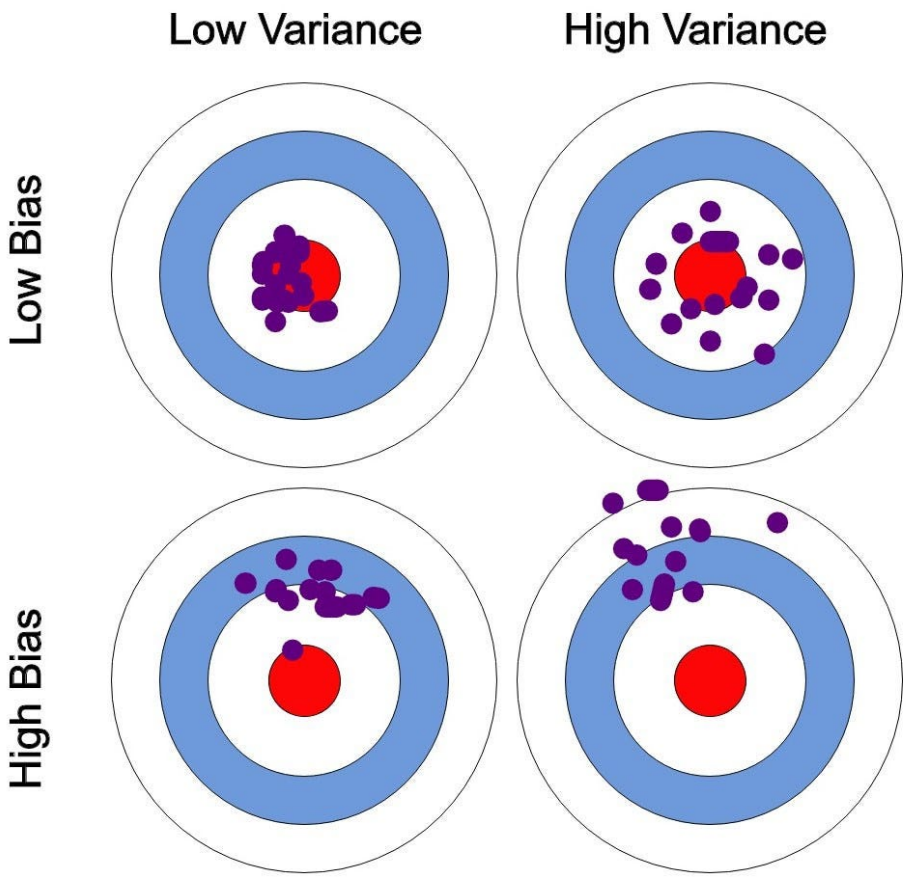
Bias

Universiteit Leiden
The Netherlands

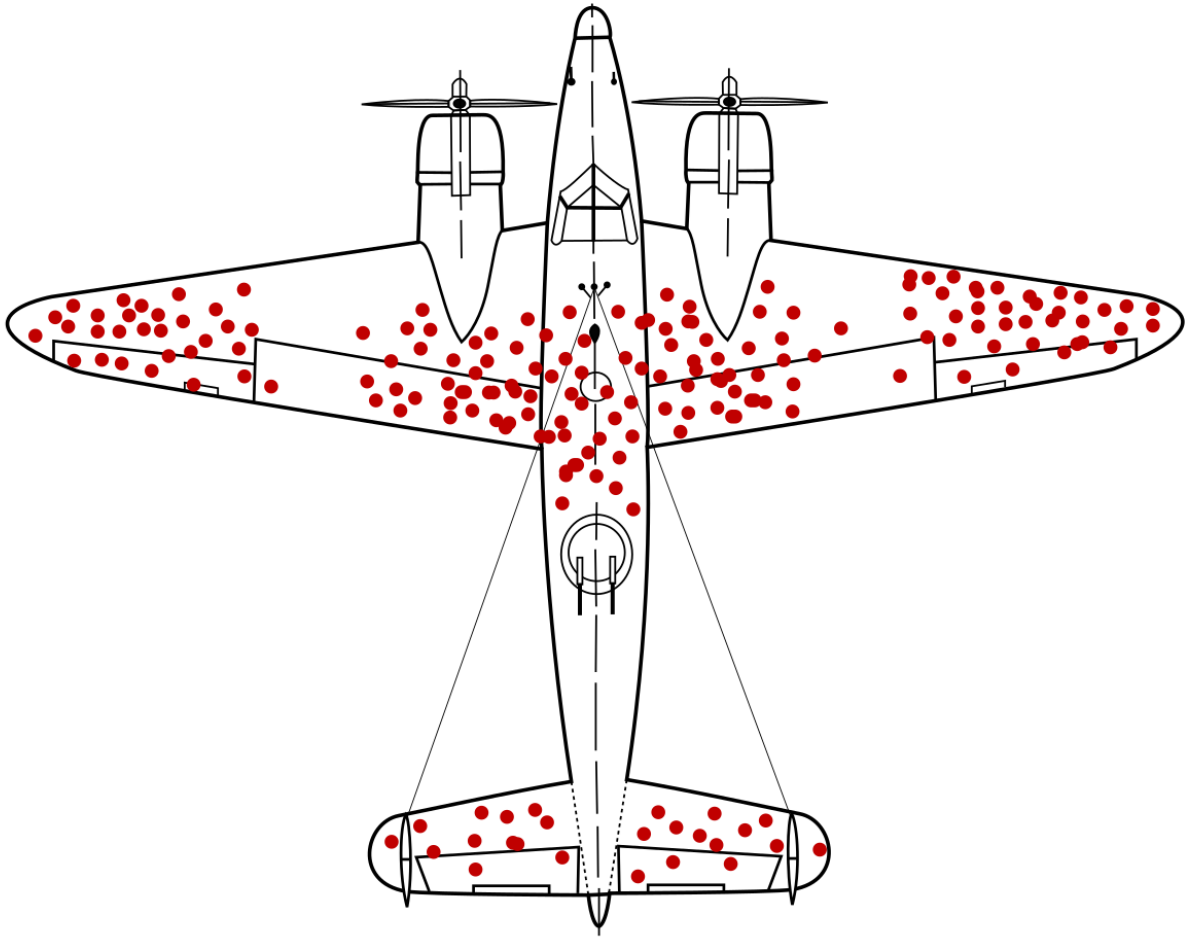
Bias

- Inclination or prejudice for or against one person or group, especially in a way considered to be unfair

- A **systematic distortion** of a statistical result due to a factor not allowed for in its derivation



Data collection!



Sampling bias

- Not all subgroups of the population are well-represented
- Sample is not suitable for inference to population

- Undercoverage

- Example: asking students in class how often they attend lectures

Nonresponse bias

- The probability of being included in the sample is dependent on certain factors

- Self-selection bias

- Example: survey to managers regarding workload
  - Very busy managers are less likely to respond: sample underestimates workload
  - Very busy managers are more likely to respond: sample overestimates workload

Response bias

- People responding falsely or inaccurately to survey questions

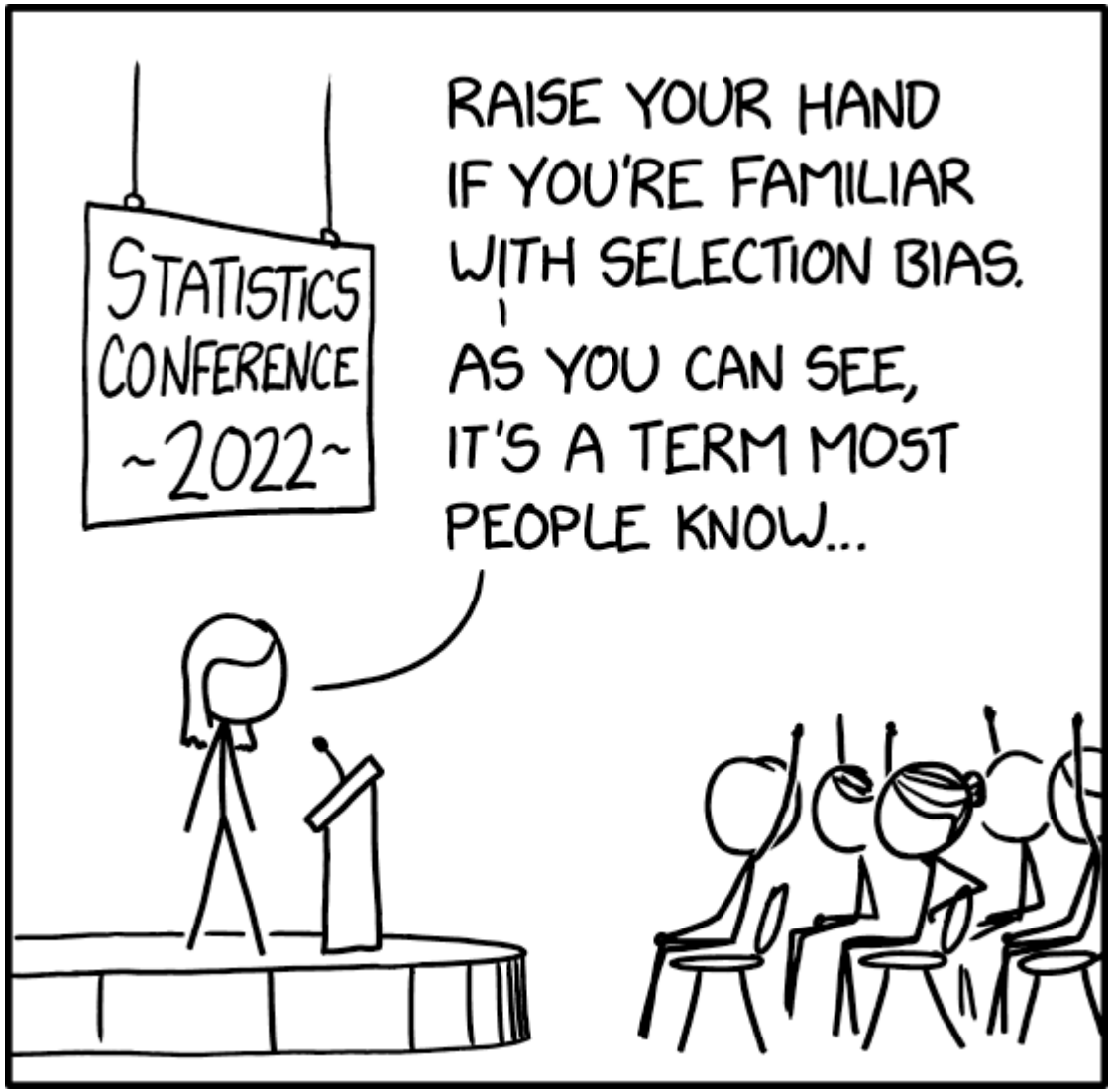
- Social desirability bias
- Courtesy bias

- Response bias: concerns subjects included in the sample
- Non-response bias: concerns missing subjects from the sample

Wrapping up

- Tutorial:
- First JASP exercise deadline 5 October

- Study chapter 4 in the book
- Brush up on probability if necessary
  - Before October 10
- No class next week!



https://xkcd.com/2618/